

**Project: Student Performance Analytics & Decision Support System**

**Client: Mega Central High School**

**Document Version: 1.0**

**Prepared By: Business Analyst**

**Document Type: Dashboard Requirements Document (DRD)**

**1. Purpose**

The purpose of this document is to define the business and functional requirements for the Power BI dashboards developed for Mega Central High School.

The dashboards provide interactive analytics to help school management, teachers and academic coordinators monitor student performance, identify learning gaps, track academic risks, and make data-driven decisions.

**2. Business Objectives**

The dashboards should enable stakeholders to:

- Monitor overall school performance
- Track subject-wise academic performance
- Identify weak students
- Monitor high-risk students
- Analyze chapter-wise learning gaps
- Compare performance across subjects
- Support intervention planning
- Improve pass percentage
- Improve student learning outcomes

**3. Stakeholders**

| Stakeholder          | Responsibility                      |
|----------------------|-------------------------------------|
| Principal            | School performance monitoring       |
| Vice Principal       | Academic monitoring                 |
| Teachers             | Student progress tracking           |
| Academic Coordinator | Curriculum improvement              |
| Business Analyst     | Requirement gathering and reporting |
| School Management    | Strategic decision making           |

**4. Dashboard Overview**

The solution consists of four interactive dashboards.

## **Dashboard 1: Executive Performance Overview**

Purpose: Provides an overall snapshot of school performance.

Target Users

- Principal
- School Management
- Vice Principal

KPIs

- Total Students
- Tests Conducted
- Average Score
- Pass Percentage
- High Risk Students
- Top Performers

Visuals

### **1. KPI Cards**

Displays

- Total Students
- Tests Conducted
- Average Score
- Pass Percentage
- High Risk Students
- Top Performers

### **2. Subject Performance**

Visual: Horizontal Bar Chart

Shows: Average percentage by subject.

Measures: Average Percentage

Axis: Subject

### **3. Student Performance Distribution**

Visual: Donut Chart

Shows: Distribution of students across performance bands.

Categories

- Top Performer
- High Performer

- Average
- Needs Support
- At Risk

#### **4. Overall Student Risk Profile**

Visual: Donut Chart

Shows: Distribution of

- Low Risk
- Medium Risk
- High Risk

#### **5. Monthly Academic Performance Trend**

Visual: Line Chart

Shows: Average monthly performance.

Includes: Target Line 70%

Filters: None

Dashboard intended to show overall school performance.

### **Dashboard 2: Individual Student Performance Analysis**

Purpose: Provides complete academic history for a selected student.

Target Users

- Teachers
- Parents
- Academic Coordinators

Filters: Student Name

Single Select Dropdown

#### **KPIs**

After selecting a student

Display

- Student Name
- Grade
- Risk Level
- Performance Band

- Average Score

## Visuals

### 1. Subject Performance Analysis

Visual: Clustered Column Chart

Shows: Average percentage in

- English
- Mathematics
- Science
- Social

### 2. Chapter-wise Learning Gaps

Visual: Horizontal Bar Chart

Displays: Bottom five chapters

Based on: Average Accuracy

Purpose: Identify weak chapters.

### 3. Assessment History

Visual: Table

Columns

- Test ID
- Date
- Subject
- Test Type
- Score Percentage
- Grade
- Pass/Fail

### 4. Student Performance Trend

Visual: Line Chart

Shows: Monthly average performance.

Includes: Target Line 70%

Interactions: Changing student updates

- KPI Cards
- Charts
- Table

### **Dashboard 3: Risk Analysis Dashboard**

Purpose: Helps identify academically at-risk students.

Target Users

- Principal
- Academic Coordinators
- Teachers

KPI Cards

- Total Students
- High Risk Students
- Medium Risk Students
- Low Risk Students
- Average Score
- Students Needing Academic Support

Visuals

#### **1. Overall Student Risk Profile**

Visual: Donut Chart

Shows: Distribution of

- Low Risk
- Medium Risk
- High Risk

#### **2. Attendance Category Distribution**

Visual: Horizontal Bar Chart

Displays: Students grouped into

- Excellent
- Good
- Average
- Poor

### 3. High Risk Students List

Visual: Table

Columns

- Student Name
- Grade
- Risk Level
- Average Percentage
- Attendance Category

Purpose: Quick identification of students needing intervention.

### 4. Subject-wise Risk Assessment

Visual: Clustered Bar Chart

Displays: Pass vs Fail percentage

For

- English
- Mathematics
- Science
- Social

## Dashboard 4: Subject & Chapter Performance Analysis

Purpose: Provides curriculum-level insights for improving academic outcomes.

Target Users

- Subject Coordinators
- Teachers
- Academic Heads

KPI Cards

- Total Subjects
- Total Chapters
- Average Accuracy
- Chapters Below Target

Visuals

#### 1. Subject-wise Performance

Visual: Horizontal Bar Chart

Shows: Average accuracy by subject.

## 2. Question Difficulty Distribution

Visual: Stacked Column Chart

Displays: Questions by

- Easy
- Medium
- Hard

Broken down by subject.

## 3. Chapters Requiring Attention

Visual: Horizontal Bar Chart

Shows: Bottom 10 chapters

Based on: Average Accuracy

## 4. Chapter Accuracy Distribution

Visual: Horizontal Bar Chart

Displays: Accuracy for weakest chapters.

## 5. Chapters Requiring Attention Table

Columns

- Chapter
- Subject
- Average Accuracy

Purpose: Provide detailed information for academic planning.

## **5. Business Rules**

### **Student Risk Level**

| Average Score | Risk Level  |
|---------------|-------------|
| $\geq 70$     | Low Risk    |
| 50–69         | Medium Risk |
| $< 50$        | High Risk   |

### Performance Band

| Average Score | Band           |
|---------------|----------------|
| $\geq 85$     | Top Performer  |
| 70–84         | High Performer |
| 50–69         | Average        |
| 35–49         | Needs Support  |
| $< 35$        | At Risk        |

### Pass Criteria

Student passes if  
Score Percentage  $\geq 35\%$

### Weak Chapter

Chapter Accuracy  $< 60\%$

Chapter Below Target

Average Accuracy  $< 60\%$

## **6. Dashboard Navigation**

| Dashboard                               | Purpose                  |
|-----------------------------------------|--------------------------|
| Executive Performance Overview          | School-level KPIs        |
| Individual Student Performance Analysis | Student-level insights   |
| Risk Analysis Dashboard                 | Academic risk monitoring |
| Subject & Chapter Performance Analysis  | Curriculum analysis      |

## **7. Functional Requirements**

The dashboards shall:

- Display real-time KPIs.
- Allow student selection using a slicer.
- Automatically refresh all visuals based on filters.
- Display monthly performance trends.
- Highlight weak chapters.
- Identify high-risk students.



- Support drill-down analysis.
- Provide printable reports.

## **8. Non-Functional Requirements**

### Performance

- Dashboard loads within 5 seconds.
- Visual interactions respond within 2 seconds.

### Usability

- Simple layout.
- Consistent colors.
- Clear labels.
- Interactive visuals.

### Reliability

- Accurate calculations.
- No duplicate records.
- Data validation completed before reporting.

### Scalability

Dashboard should support future expansion to:

- Multiple schools
- Additional grades
- More subjects
- New assessment types

## **9. Assumptions**

- Assessment data is complete.
- Student IDs are unique.
- Question bank is maintained correctly.
- Chapter mapping is accurate.
- Attendance data is available.
- Data is refreshed before reporting.

## **10. Out of Scope**

The following are not included in the current version:

- Predictive analytics
- AI-based student recommendations
- Parent portal
- Teacher portal
- Mobile application
- Automated email alerts
- Real-time attendance integration

## **11. Success Criteria**

The dashboard implementation will be considered successful if it enables stakeholders to:

- Monitor overall school performance from a single executive dashboard.
- Analyze individual student performance and assessment history.
- Identify and prioritize high-risk students for timely intervention.
- Detect subject-wise and chapter-wise learning gaps.
- Support curriculum improvements using chapter-level insights.
- Reduce manual reporting effort through automated Power BI dashboards.
- Facilitate data-driven academic decision-making across the school.